Chiral polyhedra and one projective reversal

The human_compression project

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Abstract

We recover the complete principal theorem of Elliott, Leemans and O'Reilly-Regueiro: neither $\mathrm{PSL}_2(q)$ nor $\mathrm{PGL}_2(q)$ is the full automorphism group of a chiral polyhedron. The common intertwiner theorem gives one reversal for every generating pair, including characteristic two and the small fields. We also give the precise characteristic qualification for the source's auxiliary count. The full common theory is an explicit dependency of this application.

Theorem 1. *For every prime power q, no abstract chiral polyhedron has full automorphism group* $\mathrm{PSL}_2(q)$ *or* $\mathrm{PGL}_2(q)$.

Proof. Let a, b be its distinguished rotations, with $(ab)^2 = 1$. Choose invertible matrix representatives for a and ab. Their generated projective group contains the opposite elementary unipotents in $\mathrm{PSL}_2(q)$, whose unique algebraic fixed points differ. Thus the representatives have no common invariant line even over an algebraic closure. By the simultaneous-reversal theorem of [U], their additive commutator X is invertible, X^2 is scalar, and conjugation by $[X]$ inverts both projectivities. It therefore inverts a and fixes ab, sending b to a^2b. This involutory automorphism normalizes both projective groups. It is exactly the reflection forbidden by the retained rotation-group criterion, a contradiction. The argument includes $q = 2, 3$; it does not replace these fields by an unproved computer lookup. □

The matrices underlying this theorem are fully explicit: $X = AB - BA$ satisfies $XA = \mathrm{adj}(A)X$ and $XB = \mathrm{adj}(B)X$. The common proof establishes both these identities and the nonzero determinant from irreducibility. No finite classification of generating pairs is needed. This replaces the source's three separate fixed-point cases (Lemmas 4.4–4.6) by a single theorem with strictly weaker hypotheses: the product relation and cyclic intersection are not needed to construct the simultaneous inverter, only to interpret it as a polyhedral reflection.

1 The point stabilizer and the auxiliary correction

We record the elementary projective-line inputs rather than retain an unstated exception. A nonidentity element whose order divides $q = p^d$ is unipotent and has order p, by its two-dimensional Jordan form. All such elements are conjugate in $\mathrm{PGL}_2(q)$ and lie in $\mathrm{PSL}_2(q)$. A noninvolutory semisimple projectivity has two rational fixed points in the split case and none in the nonsplit case; its eigenvalue ratio has order dividing $q - 1$ or $q + 1$, respectively. Conversely these divisibilities distinguish the cases since $\gcd(q-1, q+1) \leq 2$. A projectivity interchanging two distinct points is antidiagonal after sending the points to zero and infinity, and its square is scalar. These are the precise forms of source Lemma 4.2; the identity must be excluded from its first assertion.

Normalize a nonidentity unipotent to $u : x \mapsto x + 1$. An inverting projectivity fixes its unique fixed point infinity and therefore has form $x \mapsto ax + b$. The conjugation equation forces $a = -1$. In odd characteristic every $b \in F_q$ gives a nonidentity involution, so the source Lemma 4.3 count is q. In characteristic two these maps are translations: $b = 0$ gives the identity and the remaining $q - 1$ give involutions. Thus the literal unrestricted count is $q - 1$ in characteristic two. This correction changes neither the principal theorem nor the uniform reversal proof above.

2 Source and evidence boundary

The original result and its geometric approach are due to Elliott–Leemans–O'Reilly-Regueiro [ELO]. We bind their complete eight-page authored edition. Its Table 1 is explicitly borrowed from Hartley–Hubard–Leemans' atlas; that motivating census is not claimed as a new calculation here. Its Conjecture 3.1 is the principal theorem; its subsequent rank and rank-four-existence questions were addressed in Leemans–Moerenhout–O'Reilly-Regueiro, the companion application in this same topic. They are recorded historically, not counted as new answers.

The common paper [U] and the retained group–polytope correspondence are the full proof dependencies. The new Lean module certifies actual commutator intertwining and scalar-square identities. Absolute irreducibility and the projective and geometric interpretation remain written proofs; no complete Lean polyhedron theorem is claimed. There is no required finite program for this particular principal result.

References

[U] The human_compression project, *Common theory for projective markings and rotations*, current complete common paper.

[ELO] D. Elliott, D. Leemans and E. O'Reilly-Regueiro, *Chiral polyhedra and projective lines*, Int. J. Math. Stat. 16(1) (2015), 45–52; bound eight-page authored edition.

[1] E. Schulte and A. I. Weiss, *Chiral polytopes*, Applied Geometry and Discrete Mathematics, DIMACS 4 (1991), 435–516; retained rotation correspondence.